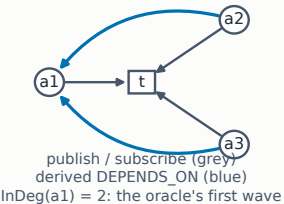
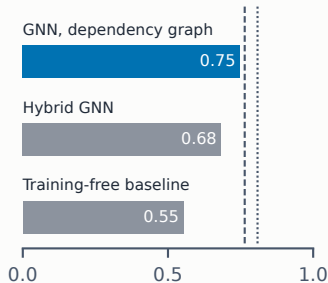


1. Derive dependencies



2. Predictors vs. references



Spearman ρ , 12 synthetic architectures
lines: references InDeg 0.76, first-order I^* 0.81

3. Beyond one simulator

Dependency counts restate the simulator: references, not predictors.

Registered primary contrast null;
hybrids beat baseline, not base learners

Queue-flow surrogate pays:
GBM 0.80 vs 0.71; GNN fails (0.60).

80% of the critical set needs
the top 40% by the best GNN.

One simulation: seconds; GNN
feature extraction: $\sim 5.6\times$ that.